

PRODUCTIVITY OF MICROFINANCE INSTITUTIONS AND FINANCIAL BURDEN TOWARDS THE CLIENTS: THE CASE OF BANGLADESH

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Abstract

This study aims to establish an empirical relationship between productivity performance of microfinance institutions (MFIs) and portfolio yield in the presence of intense debate between financial and social outreach goals. Using the recent data of 162 MFIs ranging from 2007- 2012 in Bangladesh by deploying Ordinary Least Square (OLS) and Generalized Methods of Moment (GMM), this study finds that financial and social outreach productivity positively increases the portfolio yield of MFI. Additionally, the number of active borrowers, administrative cost, loan loss, average savings and return on equity are also significantly positively related to portfolio yield. The policy implications are further discussed based on the empirical findings.

Key words: Portfolio yield, MFIs, Financial, Social outreach, Productivity, Bangladesh

JEL: G21, O16, L11, D22

1. Introduction

In recent years, there have been an increasing amount of studies on productivity assessment of microfinance institutions (MFIs) globally (Azad, Masum, Munisamy, & Sharmin, 2015; Ben Soltane, 2014; Gebremichael & Rani, 2012; Mia & Chandran, 2015; Tahir & Tahrir, 2014; Wijesiri & Meoli, 2015). Majority of the studies have shown that on average there is a progress on MFIs productivity performance (Azad et al., 2015; Ben Soltane, 2014; Gebremichael & Rani, 2012; Mia & Chandran, 2015; Tahir & Tahrir, 2014; Wijesiri & Meoli, 2015). In the institutional perspective, progress in productivity performance can be perceived as a positive accomplishment of MFIs as they are using their scarce resources to attain their objectives. However, whether the

achievement in MFIs productivity progress is causing financial burden towards their clients are worrisome not only to the welfarists but also towards the practitioners, donors and academic scholars (Basharat, Hudon, & Nawaz, 2015).

Furthermore, it should also be noted that despite decomposing the productivity index in the above studies to understand the sources of productivity growth/change, rarely they have highlighted whether this productivity progress comes with any price such as high interest rate¹. It is also evident that providing financial services to the poorest of the poor is highly costly, and hence, the services come with higher interest rate (Conning, 1999; Correa & Correa, 2009; Cull, Harten, Nishida, Rusu, & Bull, 2015). Therefore, attaining productivity progress and serving the poor community simultaneously has attracted attention aimed at understanding its dynamism. However, whether such advancement in microfinance productivity performance has any impact on their portfolio yield have not been fully explored. Thus, the goal of this paper is to investigate whether there exist a possible empirical relationship between portfolio yield and productivity.

Microfinance programs are largely used for financial inclusion of poor communities in most of the developing countries which has recently further extended to some developed countries as well to alleviate poverty (Fernando, 2006; Vigenina & Kritikos, 2004). Due to the weak formal lending institutions in most of the developing countries, microfinance is compensating such credit gap in the overall market although sometimes the funds are coming through debt financing and ignoring investment in other sectors (Cassimon & Vaessen, 2007; Estrin & Prevezzer, 2010). However, the sector is intense in the developing countries and the sectorial nominal interest rate ranges between 20% to 80% and most of the cases well above 30% (Armendáriz & Morduch, 2010; Cull, Demirgüç-Kunt, & Morduch, 2009; Helms & Reille, 2004). Evidence suggest that some of the MFIs were forced to close down their operations due to high interest rate in their loans (Basharat et al., 2015).

The impact of high interest rate in microfinance operation has bedeviled the concept and created severe impact towards borrowers. The adverse impact of high interest rate on borrowers livelihoods has been well documented (Book, 2011; Gaiha & Kulkarni, 2013; Khan & Rahaman, 2007; Kodongo & Kendi, 2013; Rosenberg, Gonzalez, & Narain, 2009; Sun & Im, 2015).

¹ Interest rate and portfolio yield are interchangeably used throughout the article.

Nonetheless, academic scholars also have given special focus to investigate the factors behind such high interest rates. Among factors identified are: cost of funds, loan tenure, loan amount, risk, collateral, gender, institutional status and lastly the profit making intention of the financier and institutions (Dorfleitner, Leidl, Priberny, & von Mosch, 2013; Fernando, 2006; Ghosh, 2013; Roberts, 2013). Goodwin-Groen (2002) argued that MFIs should lower down their interest rate by enhancing their efficiency not merely by subsidization from donors as it will distort the overall market and increase rent seeking activities. Recently, Basharat et al. (2015) empirically examined the efficiency and interest rate. However, to the best of the author's knowledge, how the productivity performance of MFIs can affect interest rate have not been examined yet.

This study differs from Basharat et al. (2015) and perhaps from other studies in the filed in few different aspects. Firstly, Basharat et al. (2015) have linked the efficiency of MFIs with interest rate while this study uses productivity concept. Literally and practically, there are subtle differences between these two concepts although they are very closely related to each other (Al-Darrab, 2000). A comprehensive discussion and relationship between productivity and efficiency can be found in Al-Darrab (2000). Secondly, there are methodological differences between Basharat et al. (2015) and this study. They have used Data Envelopment Analysis (DEA) efficiency scores while this study uses Malmquist total factor productivity approach for the first stage analysis. The Malmquist index is an extension of DEA. In the second stage analysis, they have relied only to Tobit regression while this study uses Ordinary Least Square (OLS) as a benchmark and Generalized Methods of Moment (GMM) to see the robustness of the estimation. There are also variations in choosing the inputs and outputs and variables for regression analysis. This study further includes some variables (loan loss, return on equity, active borrowers, and administrative expense) which are significantly important but have not considered in previous studies (Basharat et al. (2015)). They have used data ranging from 2005 to 2008 while this current study uses recent data ranging from 2007 to 2012, and the time period is also longer in this study.

Thirdly, this study considers a single country, which is Bangladesh. Although the concept of microfinance are quite homogeneous across the world, but their operation and performance may be heterogeneous due to the differences in socio-economic and other factors (Ahlin, Lin, & Maio, 2011). Lastly, this study is the extension of previous empirical work by Mia and Chandran (2015), where the authors have estimated the productivity of MFIs by categorizing the output indicators

into financial and social outreach separately. However, this study will further establish the empirical relationship between productivity and interest rate. The contribution of this study is obvious as the resulting outcomes could serve as guidelines to microfinance operation for sustainable development of the sector.

The organization of this study is as follows. Section 2 briefly discusses the existing literature on productivity and interest rate followed by deriving a conceptual framework. Section 3 explains the methodology that consists of selection of inputs, outputs for assessing productivity and variables used in the regression analysis. Next, section 4 reports and discusses the empirical findings. Lastly, this study concludes with some policy implications, limitations and future research directions.

2. Productivity and interest rate: A conceptual framework

The recent investigation of MFIs productivity assessment (Ben Soltane, 2014; Gebremichael & Rani, 2012; Mia & Chandran, 2015; Tahir & Tahrim, 2014; Wijesiri & Meoli, 2015) has pointed out several important issues. If the productivity progress of MFIs really helps to lower down the interest rate and help MFIs to provide services to the hardcore poor of the society, obviously attaining productivity progress will be an expected outcome for the management of the MFIs. Economically, the productivity progress in microfinance is expected to enhance competition in the sector. Academic community has linked the productivity of firms or institutions with competitions or *vice versa* (Carayannis & Grigoroudis, 2014; Jaumandreu, 2009; Latruffe, 2010; Sanyal & Shankar, 2011). Then, the relationship further empirically examined between competition and interest rate charging in the formal banking (Bikker & Haaf, 2002; Gilbert & Zaretsky, 2003; Peria & Mody, 2004; Polemis, 2015; Short, 1979; Smirlock, 1985; Willekens & Achmadi, 2003) and microfinance (Assefa, Hermes, & Meesters, 2013; Heinen, Hamadi, & Baquero, 2013; Ly & Mason, 2012; McIntosh & Wydick, 2005). However, since microfinance is an unique financial creation and do not usually follow the conventional mechanisms (perhaps most of the cases), thus it raises concern whether the productivity progress is really creating good or bad impacts towards the borrowers.

The productivity progress varied according to countries. The average productivity progress of MFIs were 1.7%, 3.8%, 4.3%, 4.9%, and 7% per year in Cambodia, Ethiopia, Bangladesh, Middle East and North Africa, and Kenya respectively. However, Azad et al. (2015) have found that average productivity progress of major MFIs in Bangladesh was 93.5% per annum, that is

comparatively higher than other country's productivity studies in MFIs. Their high productivity progress was mainly due to the data noise and misspecification of the inputs and outputs. They have chosen two inputs, which are financial cost ratio and operating expense while number of employees as another important input was excluded. Number of employees has been used in almost all of the productivity studies in microfinance productivity assessment (Ben Soltane, 2014; Gebremichael & Rani, 2012; Mia & Chandran, 2015; Wijesiri & Meoli, 2015). Moreover, the two outputs chosen in their analysis rarely represent the dual objectives (financial and social outreach) of MFIs. Nonetheless, Azad et al. (2015) did not adduce reasons for such high growth rate in their discussions.

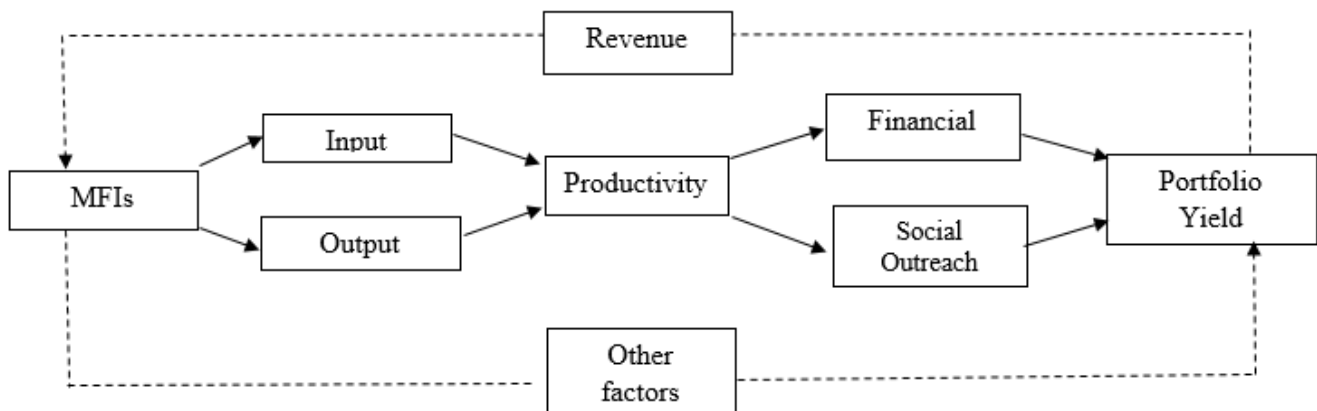
After decomposing the total factor productivity index (TFP), most of the researchers linked overall productivity progress to improvement in MFI management practices cum technical efficiency change (TEC) (Ben Soltane, 2014; Gebremichael & Rani, 2012; Mia & Chandran, 2015). As the sector is leading to its maturity and MFIs 'know-how' techniques have largely benefited their productivity performance (Mia & Chandran, 2015). Previous studies also evident that most of the MFIs have observed deterioration in the technological improvement (Ben Soltane, 2014; Gebremichael & Rani, 2012). In contrast, Mia and Chandran (2015) found that productivity progress was accelerated by technological change when separately estimated for financial productivity. This echoes the findings of Wijesiri and Meoli (2015) and Tahir and Tahrir (2014)².

Until recently, interest rate in microfinance is still one of the main factors that have derided the fame of microfinance from hero to zero as a tool to alleviate poverty. Nonetheless, high interest in microfinance operation has created concern among policy makers and welfarists across Asia and Pacific. Nobel Laureate Muhammad Yunus, the father of modern microfinance has heavily criticized those MFIs by calling 'new loan sharks' as they deviated from the original objective of microfinance that is providing financial services to the poorer community without profit making intention (Amy, 2010; Mitra, 2009). However, despite the argument of high interest rate for sustainability of the MFIs (Fernando, 2006; Kar & Swain, 2014), Bangladesh Micro-credit Regulatory Authority (MRA) has capped the interest rate to maximum 27% per annum in declining

² Technological improvement in microfinance may refer to new loaning methods, new products, usage of information and communication technology and etc. in the microfinance operation.

basis which is still several percent higher than the formal lending institutions (Islam, Porporato, & Waweru, 2014; Micro-Credit Regulatory Authority, 2015). Similar interest rate ceiling is practiced in South African MFIs to preserve the clients welfare (Mohane, Coetzee, & Grant, 2000). In contrast, although Reserve Bank of India has capped the microfinance interest rate on 26% per annum in 2011 (Kashyap, 2011), however, they have lifted the ban in 2014 and linked interest rate to the sources of funds to provide leeway to the lenders (The Economic Times, 2014). This study links the productivity performance of MFIs with their interest rate charging. Figure 1 shows the conceptual framework between MFIs productivity and interest rate.

Figure 1: Conceptual framework between productivity and interest rate in microfinance.



Source: Authors proposed framework.

Methodology

3.1. Ordinary least square technique (OLS)

This study has used DEAP2.1 computer program developed by Coelli (1996) to estimate Malmquist total factor productivity index. Further details of mathematical model can be found in Mia and Chandran (2015). Although Basharat et al. (2015) have used Tobit regression with DEA score in the second stage assessment, but this method is not suitable in this study. The DEA efficiency scores were in between 0 and 1 in Basharat et al. (2015), while the productivity scores are greater than 0 and exceeds 1. If the TFP value is less than 1, it reads as the productivity regress while value greater than 1, it reads as productivity progress. Hence, this study uses Ordinary Least

Square (OLS) technique as a benchmark while Generalized Methods of Moment (GMM) is used to ascertain the results.

The OLS is one of the oldest statistical technique still use in the modern economic literature (Abdi, 2007; Pohlman & Leitner, 2003). The idea of OLS technique was first introduced by Gauss in 1794 (Abdi, 2007). This method is largely used in the formal banking and microfinance literature due to several properties (Adeusi & Olanrewaju, 2012; Agbolade, 2011; Awdeh, 2012; Cull et al., 2015; Quayes, 2012, 2015; Sufian, 2007; Sufian & Shah Habibullah, 2010). The OLS shows the relationship between dependent and independent variables where there is a linear relationship existing between them. The data pattern supports to use OLS as possible linear relationship between dependent and independent variables have been observed³. In addition, this study also found that there is no severe correlation among the independent variables that supports another important assumption of OLS (see table 3). Thus, this study uses the OLS, and the empirical expression can be derived as follows;

$$LnPY_{it} = \alpha_0 + \alpha_1 LnNAB_{it} + \alpha_2 LnASAV_{it} + \alpha_3 LnADM/ASSET_{it} + \alpha_4 LOA_{it} + \alpha_5 ROE_{it} + \alpha_6 LnFTFP_{it} + \alpha_7 LnSTFP_{it} + \varepsilon_{it} \quad (1)$$

Where, α_0 is the constant term and α_1 to α_7 are the coefficients of natural logarithm of number of active borrowers, natural logarithm of average savings, administrative expense, loan loss, return on equity, natural logarithm of financial total factor productivity and natural logarithm of social outreach total factor productivity of MFIs respectively. Basharat et al. (2015) and Sufian (2011) also log transformed efficiency and productivity score respectively in their empirical analysis. Lastly, i denotes the individual MFIs and t is the time period while ε is the stochastic error term.

As a robustness check, this study uses Generalized Methods of Moment (GMM) to see the consistency of the coefficients of the independent variables estimated under OLS. The fundamental assumption in a regression model shows that independent variables are uncorrelated with the error/disturbance term (Ajisafe & Akinlo, 2013). However, if this assumption is violated, the OLS estimation is biased and inconsistent and do not hold the Best Linear Unbiased Estimation (BLUE) properties. In that case, GMM is a suitable model that can check for heteroskedasticity and serial

³ The graphical representation of data can be obtained from the corresponding author.

correlation between the exogenous variables. Furthermore, this study adopted GMM estimator proposed by Arellano and Bond (1991) that eliminates possible endogeneity.

$$LnPY_{it} = \beta_0 + \beta_1 LnPY_{it-1} + \beta_2 LnNAB_{it} + \beta_3 LnASAV_{it} + \beta_4 LnADM/ASSET_{it} + \beta_5 LOA_{it} + \beta_6 ROE_{it} + \beta_7 LnFTFP_{it} + \beta_8 LnSTFP_{it} + \varepsilon_{it} \quad (2)$$

$LnPY_{it-1}$ is the first year lag period of portfolio yield.

3.2. Choosing input, output and variables

The selection of inputs, outputs and variables for productivity and regression analysis are very important for analyzing MFIs performance. This study adopted two stage analysis, first stage estimates productivity index followed by a regression analysis with set of explanatory variables including productivity index against portfolio yield. A comprehensive discussion and rationale behind choosing inputs and outputs in estimating productivity can be found in Mia and Chandran (2015). In choosing the variables for regression analysis, the existing literature have been used as benchmark.

The number of active borrowers (NAB) can be used to measure the demand of microfinance loans. Assuming all other things remain same or *ceteris paribus*, a higher demand for loan is expected to increase the portfolio yield under the concept of market demand and market supply. MFIs also incorporating several other services such as savings, remittance, micro-insurance and etc. Initially, most of the MFIs in developing countries were receiving significant amount of donations and concession loans from multinational development agencies and from the domestic government sources as well. However, with the advent of microfinance operation, quite large amount of outstanding loans are financed by borrowers savings (Micro-Credit Regulatory Authority, 2015) . As the MFIs also need to provide the savers market ongoing interest rate after the subtraction of their administrative cost, a higher average saving is expected to increase the portfolio yield assuming these savings are used to generate loans.

Researchers are unanimous that MFIs charge high interest rate due to their large administrative expenses, it is one of the main reasons why formal banking institutions fail to provide the financial services to the poor (Basharat et al., 2015; Mia, 2014). The microfinance sector remains a labor intensive industry (Mia, Nasrin, & Cheng, 2015) managing short term, small size of loans are costly (Ghosh, 2013). Hence, it is reasonable to expect a positive relationship between administrative expenses and portfolio yield. The recent trend in microfinance operation is to achieve financial sustainability. Hence, sometimes the MFIs are even forced by their stakeholders/shareholders to charge interest rate in a way that can cover up their total cost.

This study assumes that the return on equity (from the stakeholders/shareholders and retained earnings) is reinvested in microfinance operation as loan outstanding. Theoretically, more return on equity accumulatively increases the ownership of stakeholders/shareholders in the MFIs, and they would expect to have better return from their money invested into the business. Hence, this study expects a positive relationship between return on equity and portfolio yield. Loan loss is estimated by impairment allowances divided by portfolio at risk greater than 30 days and expected to have positive relation with portfolio yield. When the impairment allowance increases, the credit quality of the loan portfolio deteriorate (Bruett et al., 2005). So as a compensation of high risk involved in the loan portfolio and maintain the steady revenue income, the portfolio interest rate goes up.

The two main independent variables in this study, financial total factor productivity (FTFP) and social outreach total factor productivity (STFP) estimated by using Malmquist index approach are expected to have positive impact towards portfolio yield. The main argument behind this relationship is predicated on the fact that the interest rates charged by MFIs is mainly for them to remain productive, assuming they are rationale. A significant percentage of MFIs interest are used to settle employment benefits (salaries, pensions, fringe etc.), training and other associated human development costs. Hence, enhancing the financial productivity comes with higher portfolio yield assuming that the interest is the only available source of MFIs income. Furthermore, Conning (1999) and Correa and Correa (2009) showed that outreaching to the poorer community is costly and services can be provided only with high interest rate. Since this study uses average loan outstanding and number of savers to measure social outreach productivity estimation, the greater depth (average loans) and breadth (savers) needs to be associated with high cost. Although high

interest rate is a burden towards borrowers, but this might be the trade-off between the financial services and cost.

Lastly, Portfolio yield is one of the good measurement to capture the interest rate from the vantage point of the supplier (Rosenberg, Gaul, Ford, & Tomilova, 2013). Although annual percentage rate (APR) paid by the borrowers could have been a better choice, but data availability limit the use of such variable. However, there is a distinctive differences between financial revenue/assets (FINREV) used to estimate financial productivity and portfolio yield in regression analysis. Although lion share of the MFIs earnings are coming from portfolio yield, FINREV simply shows the ability to generate revenue from asset held, while PY is the income generated from loan outstanding. Table 1 provides the definition of variables as well as reports the previous studies that have used those indicators.

Table 1: Selection of input and output

Input/output	Definition	Used in the previous studies.
Number of employees (<i>NUEMP</i>)	Total number of staff currently working. (number)	Ben Soltane (2014);Mia (2014);Mia and Chandran (2015);Mia et al. (2015);Gutierrez-Nieto, Serrano-Cinca, and Mar Molinero (2007); Gebremichael and Rani (2012)
Operating exp./Asset (<i>OPEX.</i>)	Expenses related to operations, including all personnel expense, depreciation, amortization, and administrative expense over average total assets (%).	Gutierrez-Nieto, Serrano-Cinca, and Mar Molinero (2007); Mia and Chandran (2015); Ben Soltane (2014);Worthington (1998);Berger and Humphrey (1997);Gebremichael and Rani (2012)
Financial Revenues/Assets (<i>FINREV</i>)	Total Financial revenue earned including interest, fees and commissions on gross loan and financial assets over average total assets (%).	Gutierrez-Nieto, Serrano-Cinca, and Mar Molinero (2007); Mia and Chandran (2015); Ben Soltane (2014); Mia et al. (2015); Gebremichael and Rani (2012)
Average loan balance per borrowers (<i>AVG.LOAN</i>)	Gross loan portfolio divided by total number of active borrowers (Taka).	Hisako (2009); Quayes (2012); Mia and Chandran (2015);Mia et al. (2015);Louis, Seret, and Baesens (2013).
Number of savers (<i>NUSAV</i>)	Total number of savers (number).	Mia and Chandran (2015);Lafourcade, Isern, Mwangi, and Brown (2005);Yaron (1994)
Variables used in Regression Analysis		
Average Savings (<i>ASAV</i>)	Average savings balance per member (Taka).	Basharat et al. (2015);De Silva (2012); Fiebig, Hannig, and Wisniwski (1999); Lafourcade et al. (2005); Hendricks and Chidiac (2011).
Number of active borrowers (<i>NAB</i>)	Total number of active borrowers currently taking loans from MFIs (number).	Twaha and Rashid (2012);Severino (2013); Zerai and Rani (2012); Rahman and Mazlan (2014).

Administrative expense/average asset (<i>ADM/ASSET</i>)	Total administrative expense plus depreciation/average asset (%).	Rosenberg et al. (2013); Mia et al. (2015); Mohane et al. (2000).
Return on equity(<i>ROE</i>)	Net operating income after taxes divided by the average equity (%).	Hartarska and Nadolnyak (2007); Tucker (2001); Schreiner (2002); Kinde (2012); Agarwal and Sinha (2010); Gutierrez-Nieto, Serrano-Cinca, and Mar Molinero (2007)
Loan loss (LOL)	Impairment loss allowance/ <i>PAR</i> * > 30 days.	Mersland and Strøm (2008); Ahlin et al. (2011); Morduch (1999b); Quayes (2015); Cull et al. (2015).
Financial total factor productivity (ftfp)	Financial TFP for MFIs (number).	Mia and Chandran (2015)
Social outreach total factor productivity (stfp)	Social TFP for MFIs (number).	Mia and Chandran (2015)
Portfolio yield (py)	Total interest income divided by average loan outstanding (%).	D'espallier, Guérin, and Mersland (2011); Gutierrez-Nieto, Serrano-Cinca, and Molinero (2007); Cull and Morduch (2007); Mersland (2009); Mia (2014)

Source: Authors compilation from various sources. * Portfolio at risk

3.3 Data and sources of data

The differences have been observed in policy environment between Bangladesh and Indian microfinance sector and perhaps with other parts of the world where microfinance programs are commonly used as a poverty alleviation tool. Nonetheless, macro-economic factors also play significant role in microfinance performance (Ahlin et al., 2011). Hence, pooling all the MFIs into single production frontier, estimating productivity/efficiency and making comparisons may not truly reflect the actual outcome⁴. Nonetheless, aggregation of data not only loses the crucial information about the context of MFIs, but also provides mix outcome (Basharat et al., 2015). Hence, only Bangladesh microfinance sector is chosen as a sample for this study. Furthermore, the distinctive socio-economic characteristics of Bangladesh microfinance sector, recent economic development and as the country of origins of microfinance is expected to provide better insights of the discussion (Mia & Chandran, 2015).

The data are collected from Micro-credit Regulatory Authority (MRA) which is the governing body of microfinance sector in Bangladesh. The MRA was established in 2006 with the decree of Bangladesh Government ‘Micro-credit Regulatory Authority Act 2006’. The setup of MRA came with the advent of microfinance success worldwide when Microcredit program was declared as

⁴ Basharat et al. (2015) have estimated the efficiency of MFIs separately for each of the region.

‘The International Year of Microcredit 2005’ by United Nations and Nobel Peace prize jointly by Muhammad Yunus and his Grameen Bank in 2006. However, the original objective of MRA is to create a conducive and healthy environment for microfinance practices across the country. To ensure the transparency and accountability of microfinance operation, MRA has several strict rules and regulations, and failing to do so will lead to the sanction of the MFIs (Micro-Credit Regulatory Authority, 2015).

The data comprises of 162 MFIs ranging from 2007-2012 in line with the previous study of Mia and Chandran (2015). The database has been opened for public since 2011 and important indicators of MFIs can be found from there (www.mra.gov.bd). Unlike other similar studies where Mixmarket database are extensively used, this study refers to MRA database as it is one of the most reliable sources of MFIs data in the context of Bangladesh. Additionally, Mixmarket database only reveals the self-reported data by MFIs and the reported sample size is small compare to the actual number⁵.

4. Findings and discussion

Table 2 provides the descriptive statistics of the inputs, outputs and variables used in this study. Interestingly, the maximum number of savers which was higher than the number of active borrower’s shows that not all the clients take loans rather a significant percentage of clients save only. It is well argued in the academic literature that lending to the poor increases administrative expense. This study also observed that maximum BDT of 0.90 were recorded as administrative expense for every BDT 1.00 asset held while the mean ADM/ASSET was 0.12. As former studies have confronted that the repayment rate of microfinance loans are almost 100% (Vigenina & Kritikos, 2004), this study also observed that mean loan loss provision was close to 1% while maximum was at 25%. Nonetheless, it is also quite interesting to remark that minimum loan loss was negative (-0.070) in some MFIs. It might be due to the fact that an improved economic environment, better asset quality management and generally less historical loan loss rate reduced the overall loss provision than initially estimated. Negative ROE simply shows that the return on invested equity are fading away the original equity. It might happens for the newly established MFIs that they incurred high cost of startup, high capital investment and less loan outstanding due

⁵In Bangladesh, there are total 742 registered MFIs with clients over 33.04 million in 2014 but only 28 MFIs were reported their data in 2012 into the Mixmarket database.

to economic slow-down or other associated reasons. This negative ROE is basically the outcome when MFIs observed net loss in their operation. However, although the Micro-credit Regulatory Authority (MRA) has capped the interest rate at 27% per annum, still some of the MFIs are charging well above that threshold level (maximum 87.32%). Thus, a strong monitoring and supervising efforts are indeed necessary from the authority. However, the sectorial mean portfolio yield was almost 4% lower than the maximum limit.

Table 2: Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
NOSAV	972	65,050.040	161,216.400	134	2,289,951
AVGLOAN	972	7411.533	4171.548	1031.690	68,503.530
FINREV	972	1.235	0.879	0.050	7.270
TEMP	972	374.497	920.725	4	11,951
OPTCOST	972	0.054	0.049	0.010	0.9
ASAV	807	2,311.465	1152.573	283.830	7817.680
NAB	810	59,214.880	153,126.300	874	1,841,171
ADM/ASSET	808	0.121	0.072	0.010	0.950
ROE	806	0.171	0.238	-2.790	2.330
LOL	801	0.014	0.023	-0.070	0.250
PY	795	23.207	6.403	1.620	87.320

Source: Author's estimation

Multicollinearity is one of the statistical phenomenon in panel data analysis whereby one or more independent variables are said to be highly correlated with each other. The presence of multicollinearity may severely bias the overall estimation of the regression and it violates one of the assumptions of OLS estimation that independent variables are not highly correlated. To see how the independent variables are correlated with each other, a pairwise correlation is reported in the table 3. The result shows that the correlations between the independent variables are relatively low and do not exceed the maximum limit of 0.80. The multicollinearity problem may arise if the correlations value exceed that limit (Kennedy, 2008). The minimum correlation coefficient observed between loan loss and return on equity at -0.574. The most striking observation emerged from the comparison was positive correlation between financial and social outreach productivity (0.338). Additionally, researchers also argued that financial and social outreach of microfinance complements each other (Quayes, 2012, 2015). Hence, this study can affirm that the regression analysis is not affected by severe multicollinearity problem.

Table 3: pairwise correlation among independent variables.

	ASAV	NAB	ADM/ASSET	ROE	LOL	FTFP	STFP
ASAV	1						
NAB	0.047	1					
ADM/ASSET	-0.078	-0.074	1				
ROE	0.049	0.006	-0.121	1			
LOL	-0.024	-0.015	0.102	-0.570	1		
FTFP	-0.029	-0.027	0.143	-0.271	0.147	1	
STFP	0.036	-0.037	0.058	-0.140	-0.048	0.338	1

Source: Author's estimation

In table 4, it is observed that the financial and social outreach productivity of MFIs on average progressed at a rate of 3.9% and 5% per year respectively. Interestingly, the social outreach productivity was slightly higher than the financial. Despite the trade-off between the microfinance financial sustainability and social outreach (Cull et al., 2009; Cull & Morduch, 2007; Mia, 2014; Olivares-Polanco, 2005; Serrano-Cinca & Gutiérrez-Nieto, 2014), this study suggests that there is no tradeoff when financial and social outreach productivity are estimated separately. This finding corroborates previous studies which conclude that there is no/ neutral relationship between financial and social outreach performance of MFIs (Adhikary & Papachristou, 2014; Ben Soltane, 2012; Lensink, Meesters, & Hermes, 2011; Mersland & Strøm, 2010; Quayes, 2012, 2015; Zeraï & Rani, 2012). Hence, it suggested that a progressive productivity can be attained simultaneously both in financial and social outreach given that MFIs are using their resources efficiently.

Table 4: Financial and Social Outreach Productivity of MFIs (2007-2012).

Year	Technical Efficiency Change (TEC)	Technological Change(TC)	Pure Technical Efficiency Change (PTE)	Scale Efficiency (SE)	Total Factor Productivity (TFP)
Financial Productivity					
2007-2008	0.551	1.835	0.945	0.583	1.011
2008-2009	1.532	0.671	1.148	1.334	1.027
2009-2010	0.748	1.271	0.855	0.875	0.951
2010-2011	1.098	0.827	1.112	0.987	0.908
2011-2012	0.983	1.372	1.005	0.979	1.35
Mean	0.926	1.122	1.007	0.92	1.039
Social Outreach Productivity					
2007-2008	1.074	0.974	0.891	1.205	1.046
2008-2009	1.047	1.004	1.09	0.96	1.051
2009-2010	1.127	0.94	1.128	0.999	1.059

2010-2011	1.1	0.912	1.063	1.035	1.003
2011-2012	0.957	1.139	0.998	0.959	1.091
Mean	1.059	0.991	1.031	1.028	1.05

Source: Adapted from Mia and Chandran (2015). Note: For every MFIs productivity performance, please see the supplementary materials provided by Mia and Chandran (2015).

Although the sample size of this study consists of 162 MFIs, it is apparent from Table 5 that slightly lower than two-third of the MFIs have attained financial productivity progress while the number reached almost three quarter for social outreach productivity. However, there are some MFIs that have not observed productivity progress neither in financial nor in social outreach. Hence, those MFIs may have learned from those successful MFIs to enhance their performance in both dimensions.

Table 5: Number of MFIs recorded positive and negative TFP change (average TFP change).

	Financial Productivity	Social Outreach Productivity
TFP $\geq$ 1	101(62.35%)	119(73.46%)
TFP<1	61(37.65%)	43(26.54%)
Total MFI	162	162

Source: Estimated from Mia and Chandran (2015)

Table 6 demonstrates the impact of financial and social outreach productivity in portfolio yield estimated by OLS. The financial total factor productivity (FTFP) and social outreach total factor productivity (STFP) are separately introduced in model 1 and model 2 respectively. Both of the models have R^2 of 0.2 and statically shows that 20% of the variations in portfolio yield (PY) can be explained by the included independent variables in the model. Furthermore, the F statistics are significant for all conventional levels of significance (1%, 5% and 10%) and showed that both of the models are fit in this context.

The NAB represents the demand of credit from microfinance borrowers, it is positively correlated to portfolio yield. This finding supports the law of demand, when everything else is equal, an increase in demand will increase the price. This is true because the more NAB means the more demand of loans. Since, MFIs are bounded by the financial resources constraints, thus the PY goes up to adjust with the higher demands of loans. Saving is one of the important sources of financing for MFIs to generate loans. The average savings are quite small, require large administrative cost and sometimes come as a form of compulsory savings agreement (Al-Shami, Majid, Rashid, &

Hamid, 2013). MFIs make money by reinvesting these savings amount into loans and the difference between portfolio yield and savings/deposit rate are the actual profit for MFIs.

Nonetheless, greater savings also raise concerns to provide convenience and security of the deposits. Additionally, to provide market interest rate to the depositors also challenging in the presence of high inflation rate (Morduch, 1999a). Bangladesh economy has been hiked by average 8% per year inflation from 2007-2012 (World Bank, 2015). In this context, to offset such high inflations in savers actual return requires high nominal interest rate. This may explains the positive relationship between AVSAV and PY. Similarly, ADM/ASSET also showed positive relationship with PY as MFIs want to achieve financial sustainability by generating enough revenue to cover up the total cost. Administrative cost increases due to poor physical infrastructure and transportation particularly if the MFIs are operating in remote areas (Fernando, 2006). This finding is similar to the results of Basharat et al. (2015); Rosenberg et al. (2013); Mohane et al. (2000) that administrative cost is one of the main driving force behind microfinance high interest rate.

Table 6: Impact of financial and social outreach productivity on portfolio yield

	Dependent Variable-Portfolio Yield	
	Model-1	Model-2
LNNAB	0.014**(0.007)	0.015**(0.007)
LNASAV	0.150*** (0.023)	0.147*** (0.023)
ADMASSET	1.381*** (0.147)	1.399*** (0.146)
LOL	3.510*** (0.560)	3.686*** (0.568)
ROE	0.394*** (0.055)	0.077** (0.039)
LNFTFP	0.036* (0.019)	
LNSTFP		0.039*** (0.055)
constant	1.532*** (0.183)	1.539*** (0.183)
R2	0.202	0.203
Adjusted R2	0.197	0.197
F Statistics	32.93***	33.04***

Source: Author's estimation. Values in the parenthesis are standard error. ***, **, * represents statistical significance at 1%, 5% and 10%.

$$\text{Model-1: } \ln PY_{it} = \alpha_0 + \alpha_1 \ln NAB_{it} + \alpha_2 \ln ASAV_{it} + \alpha_3 ADM/ASSET_{it} + \alpha_4 LOL_{it} + \alpha_5 ROE_{it} + \alpha_6 \ln FTFP_{it} + \varepsilon_{it}$$

$$\text{Model-2: } \ln PY_{it} = \alpha_0 + \alpha_1 \ln NAB_{it} + \alpha_2 \ln ASAV_{it} + \alpha_3 ADM/ASSET_{it} + \alpha_4 LOL_{it} + \alpha_5 ROE_{it} + \alpha_7 \ln STFP_{it} + \varepsilon_{it}$$

LOL and ROE also showed positive impact towards PY. Economically, when the LOL provision increases, it means that the credit quality of the loan portfolio is risky and the borrowers are riskier that they might end up with loan default. Although, previously a large portion of MFI loans were basically group loans, but in recent years individual loan type is gaining more attention and superseding the former one (Lehner, 2009). Traditionally, individual borrowers are more risky than a group borrowers (Kodongo & Kendi, 2013). In a group lending methods, there is very less risk of the loan to get default (Attanasio, Augsburg, De Haas, Fitzsimons, & Harmgart, 2011). Even if it happens, other group members are subject to settle the remaining debt (Vigenina & Kritikos, 2004). So the recent trend of more individual loans severely affects loan quality and more exposure to the riskier loan portfolio normally comes with high PY. This is further compounded and aggravated by *individual multiple borrowing*⁶ phenomenon (Faruquee & Khalily, 2011; Mpogole, Mwaungulu, Mlasu, & Lubawa, 2012). This study assumes that the return on equity are reinvested into microfinance operation to expand the business. When this happens, the stakeholders/shareholders expect to have better marginal return for their investment. Thus, the ROE increases the PY as a way to enhance the shareholder/stakeholders financial interest.

Finally, returning to the expectation posed in the beginning of this study, FTFP and STFP showed positive impact towards PY. The empirical estimation showed that FTFP and STFP will lead to higher PY hence more financial burden to the borrowers. However, it does not mean that this study is opposing the view of attaining greater productivity rather it is raising concern about the productivity progress and their plausible impact towards borrowers. Moreover, this study also observed that the coefficients of STFP was slightly higher than the FTFP and STFP. There are few plausible explanations for such variations. Firstly, this study has observed average social outreach productivity (5%) greater than financial productivity (3.9%), which may have a significant impact in the regression analysis. Secondly, as the social outreach productivity means providing smaller loans to the more borrowers and attracting more savers from the available resources, it involves large amount of investment to innovate such lending technologies and diverse financial products to attract more clients. Although Bashar et al. (2015) found that only financial efficiency is positively related with interest rate, however, this study acknowledge that both SFTP and FTFP

⁶ When an individual borrowers borrows from more than one MFIs, it is called '*individual multiple borrowing*'. For more details, please see Faruquee and Khalily (2011).

are positively correlated with PY. Lastly, STFP was statistically significant for 1%, 5% and 10% while FTFP was significant for 10% only. However, both of them were significant at 5% and 10% under GMM estimation.

Our results are robust in a sense that although we have separately introduced STFP and FTFP in regression analysis, the coefficients of the variables and significance level almost remained same under models 1 and 2. Furthermore, if we consider the results of GMM estimation, this study also reconfirms the robustness of the results (see table 7). Except NAB in models 3 and 4, and ADMN/ASSET in model 3, all other variables remain significant. Thus, our study highlights that overall productivity progress in MFIs come with a higher portfolio yield regardless of financial or social outreach productivity progress.

Table 7: Impact of financial and social outreach productivity on portfolio yield- two step system GMM

	Dependent Variable-Portfolio Yield	
	Model-3	Model-4
L1. LNPY	0.100(0.064)	0.091(0.069)
LNNAB	0.010(0.013)	0.024(0.015)
LNASAV	0.122*** (0.40)	0.165*** (0.038)
ADMASSET	1.116(0.694)	1.190* (0.695)
LOL	2.498** (1.024)	2.805*** (1.044)
ROE	0.300* (0.114)	0.316*** (0.104)
LNFTFP	0.036** (0.016)	
LNSTFP		0.075** (0.036)
cons	1.557*** (0.359)	1.104*** (0.370)
AR1	0.11	0.011
AR2	0.481	0.601
Sargen Test	0.000	0.000
Hansen Test	0.151	0.109
Number of observation	624	624
Number of instruments	96	96

Source: Author's estimation. Values in the parenthesis are corrected standard error. ***, **, * represents statistical significance at 1%, 5% and 10%.

$$\text{Model-3: } \text{LnPY}_{it} = \beta_0 + \beta_1 \text{LnPY}_{it-1} + \beta_2 \text{LnNAB}_{it} + \beta_3 \text{LnASAV}_{it} + \beta_4 \text{LnADM/ASSET}_{it} + \beta_5 \text{LOL}_{it} + \beta_6 \text{ROE}_{it} + \beta_7 \text{LnFTFP}_{it} + \varepsilon_{it}$$

$$\text{Model-4: } \text{LnPY}_{it} = \beta_0 + \beta_1 \text{LnPY}_{it-1} + \beta_2 \text{LnNAB}_{it} + \beta_3 \text{LnASAV}_{it} + \beta_4 \text{LnADM/ASSET}_{it} + \beta_5 \text{LOL}_{it} + \beta_6 \text{ROE}_{it} + \beta_7 \text{LnSTFP}_{it} + \varepsilon_{it}$$

5. Conclusion

While most of the literature and policy article support the view of achieving productivity in microfinance operation, to date and best of author's knowledge, there is scant of research that address the issue of how productivity progress can affect interest rate. Hence, this study contributes to the debate of microfinance productivity performance and portfolio yield by showing the empirical relationship between them. The data were collected from MRA database including 162 MFIs ranging from 2007-2012 in Bangladesh. Malmquist index approach was used to estimate the productivity of MFIs in the first stage. Then, OLS and GMM techniques have used in regression analysis in the second stage.

This study refuted the conflicting goals between financial and social outreach productivity and these objectives seemingly be achieved through productive employment of microfinance resources. The most obvious finding from this study is that there is a positive relationship of portfolio yield with MFIs financial and social outreach productivity. This finding is supported by earlier studies that there is a price (interest rate) that need to be borne by the clients to support MFIs productivity progress over time. The expansion of microfinance programs through productivity progress has helped provide financial services to the poor and enhancing their social welfare, while the resulting productivity progress in MFIs has increased the portfolio yield. However, MFIs should distinguish between their productivity progress and financial burden to the clients to have a better impact to the society. Nonetheless, this study is not opposing the concept of attaining positive productivity rather re-assessing the impact of productivity towards MFIs borrowers. Furthermore, apart from productivity, ADM/ASSET, ASAV, NAB, LOL and ROE also showed positive impact on PY. Hence, several plausible conclusions can be drawn from here. First, MFIs should develop cost savings products and financial services that can significantly reduce interest rates. Nonetheless, to absorb the higher savings from savers, MFIs should have comprehensive savings products and they should find a niche market for investing their deposit to have better return. Since microfinance is a labor intensive industry, information and communication technology can reduce the administrative cost. Computerization of MFIs daily operation may save lion share of their administrative cost, hence reflects in a lower interest. An additional effort should be placed to lower down the loan loss provision of MFIs. Monitoring the

borrowers and comprehensive selection procedures would help much to have a better loan portfolio quality. Nonetheless, MFIs may learn and incorporate some techniques from formal banking loan portfolio management to enhance their performance. As MFIs have lack of sources of funds and increase in NAB also positively rises PY, adequate sources of funds can bridge the demand and supply gap of credits. Lastly, as microfinance programs are social movements to enhance poor peoples' livelihoods, the stakeholders/shareholders' financial interest should be critically reviewed.

The study has gone some way towards enhancing our understanding of the dynamism of microfinance productivity and portfolio yield. Although this study provides some new insights empirically, there are some inherent limitations. A comparison between different MFIs with their institutional status may help to provide additional insights of the ongoing debate. This is basically due to the lack of availability of the data. However, this study has thrown up many questions in need for further investigation. Future studies may carry out to investigate the impact of productivity in MFIs competition and innovation. Nonetheless, which combinations of productivity growth rate and portfolio yield may be complimentary and beneficial to both parties can be a milestone achievement in the future research.

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